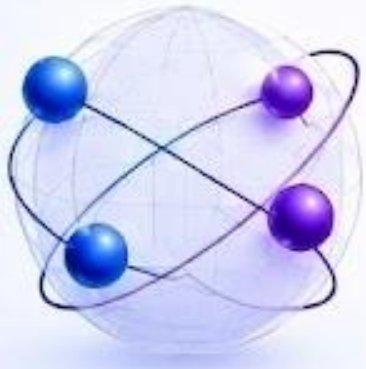
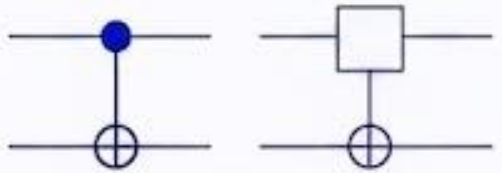
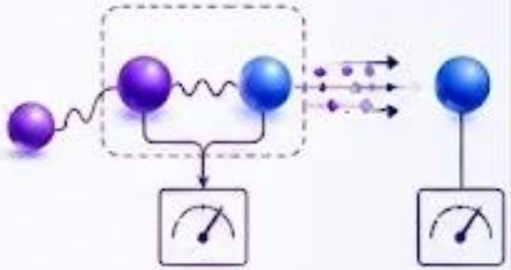
MULTI-QUBIT SYSTEMS | TENSOR PRODUCT | CONTROLLED GATES | BELL STATES | TELEPORTATION







DAY 4

Multi-Qubit Systems and Quantum Gates

Girish Belani
Project Engineer
C-DAC Hyderabad

Circuit Model of Quantum Computation

How quantum computers perform computations via qubits, quantum gates, and measurements.

Classical vs. Quantum Circuits

Feature	Classical Circuit	Quantum Circuit
Basic Unit	Classical Bits (0 or 1)	Qubits (Superposition)
Operations	Logic Gates (AND, NOT, XOR)	Quantum Gates (Unitary Matrices)
State Properties	Independent definitive states	Entanglement and Wave Interference
Output Extraction	Direct deterministic output	Probabilistic Measurement

Tensor Networks & Operations

Algebraic Approach

We can apply separable operations to entangled states, and the state can remain entangled. Consider a two-qubit state:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

Applying a Pauli-X operation on the first qubit (a NOT gate) flips it:

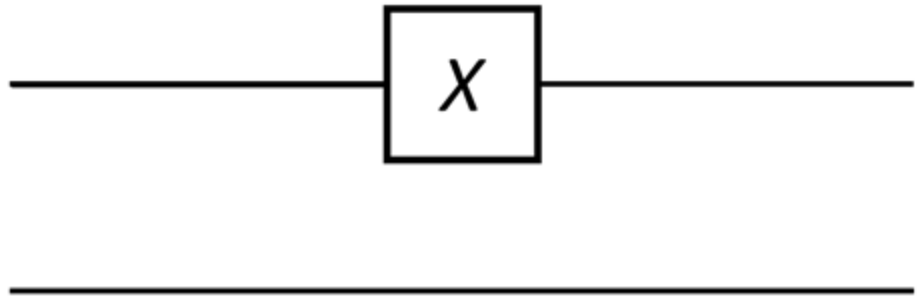
$$|\psi'\rangle = (X \otimes I)\,|\psi\rangle$$

The resulting flipped state becomes:

$$|\psi'\rangle = \alpha|10\rangle + \beta|11\rangle + \gamma|00\rangle + \delta|01\rangle$$

Tensor Network Representation

Alternatively, we can consider a tensor network representation. Each horizontal wire represents a qubit, making the visual structure of the multi-qubit system and the locality of the operation instantly intuitive.



Building Blocks: Matrices to Gates

Single-Qubit Gates

Operate on individual qubits to create superpositions or phase shifts. Common examples include Pauli-X, Y, Z, and Hadamard.

Multi-Qubit Gates

Operate across multiple qubits, fundamentally enabling entanglement. The CNOT gate is the most prominent building block.

Unitary Matrices

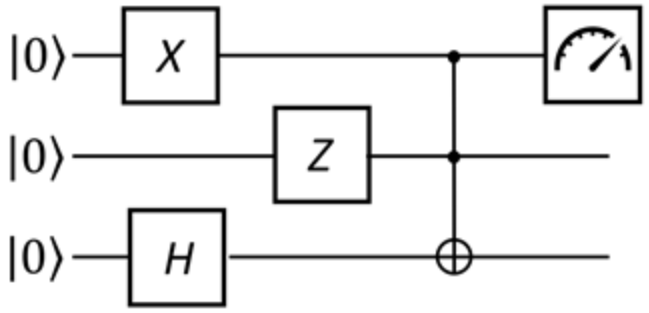
Every gate is a unitary matrix. For example, the Phase (S) gate is mathematically mapped as:

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

Quantum Circuit Lifecycle

3

Main Stages



1. Initialization

All qubits are universally prepared in a starting state, usually the zero state, ready for computation.

2. Quantum Gates

Complex unitary transformations are applied sequentially to the qubits to process quantum information.

3. Measurement

At the circuit's end, qubits are measured to collapse the quantum state and extract a final classical result.

The Matrix of a Quantum Circuit

Rule 1: Across Wires

Operations acting on different qubits simultaneously are combined using the tensor product.

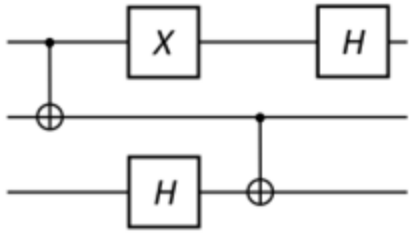
```
q_0: ——[ A ]——————
q_1: ——[ B ]——————
```

$$A \otimes B$$

This mathematical notation clearly represents operation A acting on one qubit and operation B acting on another parallel qubit.

Rule 2: Along Wires

Operations applied sequentially are combined using normal matrix multiplication, applied from right to left.



$$U = (H \otimes I \otimes I) \cdot (I \otimes CX) \cdot (X \otimes I \otimes H) \cdot (CX \otimes I)$$

Combining these matrices maps the complete unitary evolution of the entire quantum circuit into a single mathematical state.

Quantum Computational Power

> *"Any computation that exploits the quantum nature of a physical system can, in principle, be expressed using the quantum circuit model."*

Basic Structure & Initialization

The Initial State

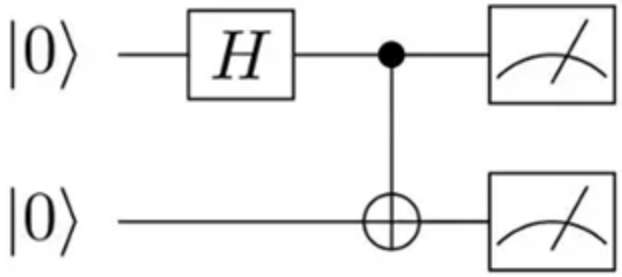
A quantum computation starts by initializing qubits. For *n* qubits, this is mathematically expressed as:

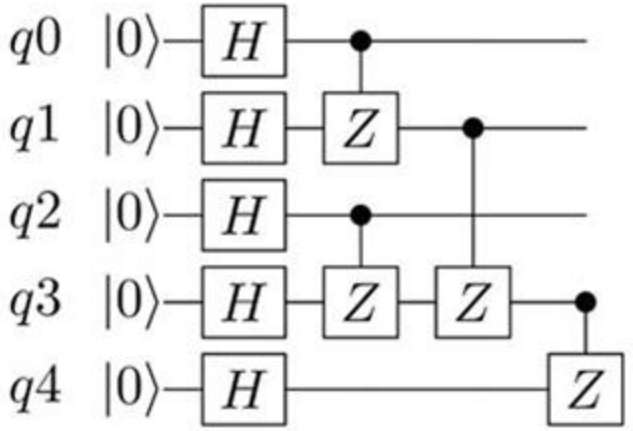
$$|0\rangle^{\otimes n}$$

Visualizing the Wires

Each horizontal line represents a qubit wire. The circuit flows chronologically from left to right.

It does not represent a physical particle traveling through space, but rather the mathematical evolution of the quantum state over time.





Apply Quantum Gates

Gate Name	Symbol	Primary Purpose
Pauli-X	`X`	Bit flip (Quantum NOT operation)
Pauli-Z	`Z`	Phase flip transformation
Hadamard	`H`	Creates an equal superposition state
Rotations	`RX, RY, RZ`	Phase and State rotation around distinct axes
Controlled-NOT	`CNOT / CX`	Two-qubit controlled operation (Enables Entanglement)

Measurement & Shots



Extracting Data

Measurement converts the fragile quantum state back into classical information (0s and 1s) allowing the results to be definitively read by standard computers.



Probabilistic Results

A Bell State yields 00 roughly 50% of the time, and 11 roughly 50% of the time. You will never observe the 01 or 10 states under ideal conditions.



Statistical Sampling

Because quantum outputs are inherently probabilistic, circuits are repeatedly executed over thousands of "shots" to construct a highly reliable data distribution.

Circuit Depth Constraints

Understanding Depth

A circuit's depth is determined by its sequential layers of operations. For example, a flow of H → X → Z on a single wire contains multiple sequential layers.

Highly complex algorithms require many layers, naturally increasing the overall **circuit depth** or **number of sequential gate layers**.

Why Depth Matters

Real quantum hardware is strictly constrained by external noise and limited coherence times. Deeper circuits provide more opportunities for quantum errors to accumulate.

Modern algorithmic designers strive to aggressively minimize unnecessary depth to maintain computation fidelity.

Real-World Quantum Computing

The Hybrid Loop

Quantum computers rarely work entirely alone. In modern hybrid models like VQE and QAOA, classical and quantum hardware collaborate closely.

01

Classical Computer

Generates algorithmic parameters and computational optimizations.

02

Quantum Hardware

Evaluates the highly complex, entangled circuit.

03

Feedback Loop

Measurement results update classical parameters iteratively to find solutions.

Locality & Physical Realization of Quantum Gates

The Locality Constraint

Mathematical Possibility ≠ Physical Feasibility

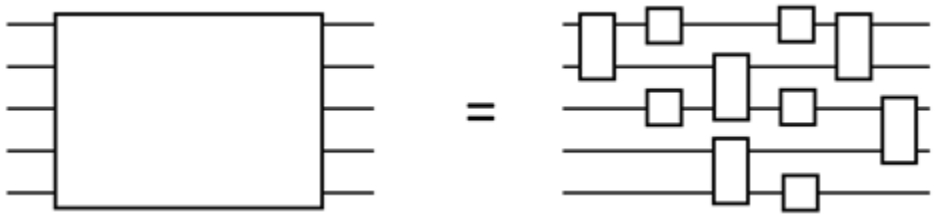
Mathematically, unitary operators can act on any number of qubits. In a physical quantum computer, however, qubits are spatially arranged and interactions are typically limited to one or two qubits at a time.

The Universal Solution

1- and 2-Qubit Gate Universality

- **Single-qubit gates:** Act on one qubit.
- **Two-qubit gates:** Create interactions between pairs of qubits, typically according to the hardware's connectivity.
- **Decomposition:** In principle, any n-qubit unitary can be decomposed into 1- and 2-qubit gates. However, for generic unitaries, the gate count can grow exponentially.

Decomposition of Arbitrary n-Qubit Unitaries



Complex quantum operations are built from simple local operations.

Local Operations ➔ Scalable Physical Implementation ➔ Universal Quantum Computation

Deriving Common Gates

Derived Gate	Construction from H & T	Mathematical Representation
S Gate (Phase)	Two sequential T gates	$S = T^2$
Z Gate (Phase Flip)	Two sequential S gates	$Z = S^2$
X Gate (Bit Flip)	Z gate sandwiched by Hadamards	$X = HZH$
Y Gate	Complex combination of X and Z	$Y = iXZ = SXSZ$

Quantum Reversibility

Equal Input & Output Dimension

Quantum gates map physical phenomena and have exactly the same number of outputs as inputs.

Unitary Matrices

Because quantum gates represent unitary matrices, they are always invertible.

Perfect Reversibility

Applying a unitary **U** can be perfectly reversed by applying its complex conjugate transpose, $U^{\dagger}$.

Reversible Circuit Representation



Inner Product in Quantum Computing

Understanding Scalars, Bra-Ket Notation, and Orthonormality

1. Converting a Ket to a Bra

Finding the Inner Product

To calculate the inner product of two states, we must first convert the starting ket into a bra by taking the complex conjugate transpose.

This operation, often denoted by the dagger symbol (†), transforms a column vector into a row vector while conjugating each entry.

The asterisk (*) signifies the complex conjugate of our complex numbers, changing the sign of the imaginary part.

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \rightarrow \langle\psi| = \begin{bmatrix} \alpha^* & \beta^* \end{bmatrix}$$

Mathematical Conversion **Ket to Bra Transition**

Key Operations:

- **Conjugate (*):** Negate the imaginary part of α and β
- **Transpose:** Flip the column vector to a row vector

Scalars Explained

1

Single Number

Inner Product = Scalar

When multiplying a bra and a ket matrix together, the vector structure disappears entirely.

The calculation yields exactly one mathematical value. Because the result is a single number, the inner product is commonly referred to as the scalar product.

Dirac Notation Origins



Ket

$$|\psi\rangle$$

A column vector representing our standard quantum state.



Bra

$$\langle\psi|$$

A row vector representing the complex conjugate transpose.

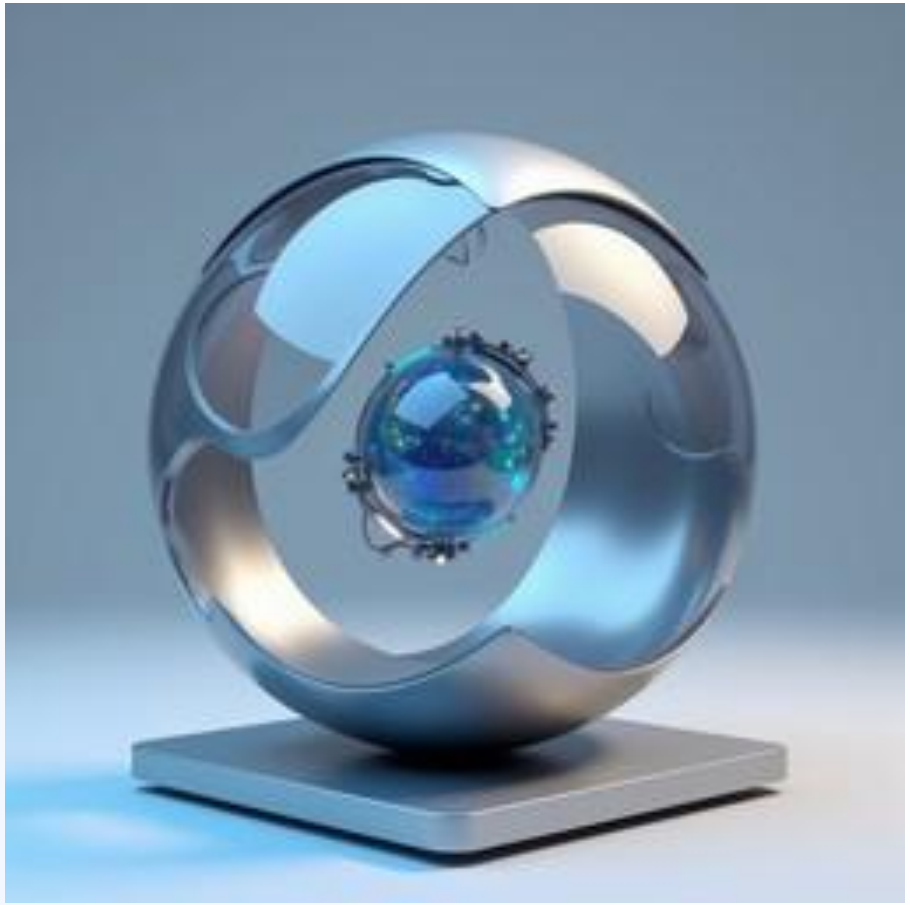


Bra Ket

$$\langle\psi|\varphi\rangle$$

Bra + Ket = Braket. The scalar result of their inner product.

Complex Conjugate Relationship



Reversing the Order

If you reverse the order of the two quantum states being multiplied, the final result automatically becomes the complex conjugate.

$$\langle \varphi | \psi \rangle = \langle \psi | \varphi \rangle^*$$

The real part stays identical, while the sign of the imaginary component flips (e.g., from +2i to -2i).

2. Orthonormality (Normalization, Orthogonality, and Valid Quantum States)

1. Normalized

$$\langle \psi | \psi \rangle = 1$$

A valid quantum state must reflect a total probability of 1. When a state's inner product with itself equals exactly one, it is considered perfectly normalized.

2. Orthogonal

$$\langle \psi | \varphi \rangle = 0$$

When two entirely different states have zero overlap, they are orthogonal. This is geometrically equivalent to operating with perpendicular vectors.

Orthogonal Basis States

A **basis** is a set of linearly independent **vectors used to represent quantum states** in a vector space.

Basis	State 1	State 2	Inner Product
Z Basis	$\|0\rangle$	$\|1\rangle$	$\langle 0 \| 1 \rangle = 0$
X Basis	$\|+\rangle$	$\|-\rangle$	$\langle + \| - \rangle = 0$
Y Basis	$\|i\rangle$	$\|-i\rangle$	$\langle i \| -i \rangle = 0$

"The inner product takes a bra and a ket, produces a scalar, and tells us about the overlap between quantum states."

The Significance of Measurement

50%

PROBABILITY OF MEASURING $|0\rangle$

Calculating Actual Probability

The magnitude squared of the *inner product* yields the physical probability of measuring a specific state.

If we measure the $|+\rangle$ state against the $|0\rangle$ basis:

$$|\langle 0|+\rangle|^2 = \frac{1}{\sqrt{2}}^2 = \frac{1}{2}$$

This fundamental mechanism links abstract mathematics to observable reality in quantum computing.

An Intuitive Picture

States	Inner Product	Physical Meaning
$\langle 0 \mid 0 \rangle$	1	Complete Overlap. The states are perfectly aligned and identical.
$\langle 1 \mid 1 \rangle$	1	Complete Overlap. The states are perfectly aligned and identical.
$\langle 0 \mid 1 \rangle$	0	No Overlap. The states are completely orthogonal (perpendicular).
$\langle 0 \mid + \rangle$	$1 / \sqrt{2}$	Partial Overlap. The states share a proportional relationship.

The Core Distinction



Probability Amplitude

$$\langle \phi | \psi \rangle$$

The raw mathematical overlap between two quantum states. Note that this value can be complex and negative, acting as an amplitude.



Actual Probability

$$| \langle \phi | \psi \rangle |^2$$

The squared magnitude. This provides the physical, real-world probability (between 0 and 1) of observing the state during measurement.

Outer Product in Quantum Computing

The Fundamental Difference

"

Inner → scalar,

Outer → matrix.

"

DIMENSIONAL OUTCOME

Inner vs. Outer Product

Suppose we have two quantum states defined as column vectors:

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \qquad |\phi\rangle = \begin{bmatrix} \gamma \\ \delta \end{bmatrix}$$

Operation	Calculation		Result
Inner Product	Row Vector × Column Vector	$\langle\psi\|\phi\rangle$	**Scalar** (single number)
Outer Product	Column Vector × Row Vector	$\|\psi\rangle\langle\phi\|$	**Matrix**

The Mathematical Difference

The Inner Product

For the inner product : $\langle \psi | \phi \rangle$

$$\begin{bmatrix} \alpha^* & \beta^* \end{bmatrix} \begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \alpha^* \gamma + \beta^* \delta$$

The result is a scalar value representing the overlap.

The Outer Product

For the outer product : $|\psi\rangle \langle \phi|$

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \begin{bmatrix} \gamma^* & \delta^* \end{bmatrix} = \begin{bmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{bmatrix}$$

The result is a fully formed matrix operator.

Example: The |0⟩ State

01

Basis State

Take the standard computational basis state as a vector.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \langle 0| = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

02

Outer Product

Construct the outer product by multiplying the state vectors.

$$|0\rangle\langle 0| = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix}$$

03

Matrix Expansion

Expand the outer product calculation into a 2x2 matrix representation.

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

Outer Product of |1⟩

01

Basis State

Applying the same mathematical methodology to the orthogonal computational basis state vector.

$$|1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \langle 1| = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

02

Outer Product

Constructing the outer product operator by multiplying the state vectors directly.

$$|1\rangle\langle 1| = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix}$$

03

Matrix Expansion

Yielding the final 2x2 complementary projector matrix representation.

$$|1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

"The outer product is especially important because it allows us to construct operators and matrices directly from quantum states."

Tensor Products in Quantum Computing

A mathematical framework for combining individual state spaces into exponentially scaling joint systems.

The Joint Quantum State

The **tensor product** (or Kronecker product) is the essential tool used to mathematically combine multiple independent qubits into a single, comprehensive joint state.

The Notation

Given single qubits **|q1⟩** and **|q2⟩**, their combined state is **|q1⟩ ⊗ |q2⟩**, often simplified to **|q1q2⟩**.

Single Qubits

A general state requires two amplitudes:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Column Vectors

In the computational basis, states are represented as column vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Why Do We Need Tensor Products?

$$2^n$$

Amplitudes required for *n* qubits

Exponential Space Creation

While a single qubit only requires 2 components (amplitudes), adding qubits multiplies the required components.

Two qubits have four possible computational-basis states ($|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$), requiring 4 amplitudes. A general two-qubit state looks like:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

The tensor product is the mathematical engine that naturally generates this exponentially scaling vector space.

The Mathematics (Tensor of Two Vectors)

The General Expansion Rule

The calculation is straightforward: multiply each element of the first vector by the entire second vector.

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a\begin{bmatrix} c \\ d \end{bmatrix} \\ b\begin{bmatrix} c \\ d \end{bmatrix} \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}$$

Example: Creating the |00⟩ State

Applying this formula, we expand two 2D column vectors representing |0⟩ into a single 4D column vector.

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1(1) \\ 1(0) \\ 0(1) \\ 0(0) \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

All Four Two-Qubit Basis States

By substituting the respective vectors for 0 and 1, we derive the mathematical definition for all computational bases.

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = |0\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|10\rangle = |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$|11\rangle = |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Tensor Product of Superpositions

Combining Two Superpositions

Suppose both qubits are perfectly distributed (e.g., after an H-gate):

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$$

Using standard FOIL distributivity rules:

$$= \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

Resulting Probabilities

Written in full column vector format:

$$|\psi\rangle = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

- Each basis state has an amplitude of **1/2**.
- The measurement probability is $|\text{Amplitude}|^2$, which is **1/4**.
- The total sum of probabilities (1/4 + 1/4 + 1/4 + 1/4) equals exactly 1.

"Multiply each element of the first vector by the entire second vector."

— The Golden Rule of the Kronecker Product

Exponential Hilbert Space Growth

A defining feature of quantum computing is how tensor products grow the dimension of the state space at an exponential rate.

Number of Qubits (n)	State-Vector Dimension (2^n)	Example Joint Basis State
1	2	$\|1\rangle$
2	4	$\|01\rangle$
3	8	$\|010\rangle = [0, 1, 0, 0, 0, 0, 0, 0]^T$
10	1,024	$\|0110010100\rangle$
30	1,073,741,824	*At this scale, standard classical RAM struggles to store the vector.* *(A 30-qubit statevector has 2^{30} amplitudes, so memory depends on precision; with complex double precision it requires about 16 GiB just for the amplitudes, before simulator overhead.)*

Tensor Products on Quantum Gates

Combining Gate Operators

Applying an **X** gate to Qubit 1 and an Identity (**I**) to Qubit 2 is represented as **X** $\otimes$ **I**.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 1\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 0\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Applying to the State

We now have a 4x4 matrix, perfectly sized to operate on a 4-dimensional two-qubit state space.

$$(X \otimes I) \mid 00 \rangle = \mid 10 \rangle$$

This matches our intuition: the X gate flipped the first bit to 1, while the Identity gate left the second bit as 0.

Distinguishing Mathematical Products

Because the names sound similar, it's crucial to understand their geometric outputs.

Operation Type	Notation	Output Type	Example
Inner Product	$\langle\psi\|\varphi\rangle$	Scalar (Number)	$\langle 0\|1\rangle = 0$
Outer Product	$\|\psi\rangle\langle\varphi\|$	Matrix (Operator)	$\|0\rangle\langle 1\|$ creates a 2x2 matrix
Tensor Product	$\|\psi\rangle \otimes \|\varphi\rangle$	Larger Vector/Space	$\|0\rangle \otimes \|1\rangle = \|01\rangle$ (A 4x1 vector)
Standard Matrix Multiplication	$\|0\rangle\|1\rangle$	*Undefined*	Cannot multiply two 2x1 column vectors.

Software Implementations

Numpy Kronecker Product

In Python, tensor products are handled using Numpy's `kron` function.

```
import numpy as np
zero = np.array([[1], [0]])
one = np.array([[0], [1]])
# Tensor product of |0> and |1>
state = np.kron(zero, one)
print(state)
# Output: [0 1 0 0] (represents |01>)
# bra_zero = zero.conj().T
```

Crucial Detail: Qubit Ordering

Mathematical Notation

Left-to-right reading (index 1 then index 2).

Software (e.g., Qiskit)

Uses *little-endian ordering*. The rightmost bit corresponds to the lowest-index qubit (q0).

Tensor Product: Combining Qubits to Entanglement

Combining Qubits

The tensor product combines the states of individual qubits into a single joint quantum state:

$|q1\rangle \otimes |q2\rangle$

Building State Space

When n qubits are combined, their tensor product forms a 2^n-dimensional state space containing all computational basis states:

$|q1\rangle \otimes |q2\rangle \otimes \ldots \otimes |qn\rangle$

Understanding Entanglement

A multi-qubit state is entangled when it cannot be expressed as a tensor product of independent individual-qubit states:

$|\psi\rangle \neq |\psi1\rangle \otimes |\psi2\rangle$

tensor products construct joint multi-qubit state spaces; interactions such as entangling gates can produce non-separable states

Foundational Controlled Gates

CX, CZ, and CCX (Toffoli) in Quantum Computing

Conditional Logic in the Quantum Realm

"

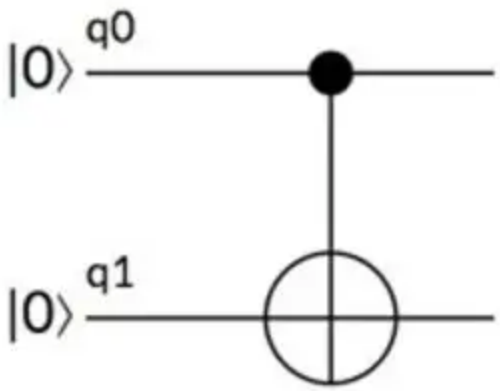
$$\text{If Control is state } |1\rangle \rightarrow \text{Apply Operation to Target}$$
$$\text{If Control is state } |0\rangle \rightarrow \text{Do Nothing}$$

— Operational Principle of Controlled Quantum Gates

1. CX Gate (Controlled-X or CNOT)

The CX gate is the cornerstone of multi-qubit interaction, effectively acting as a quantum XOR operation.

- Configuration: 1 control qubit, 1 target qubit.
- Mechanism: It applies an X (NOT) gate to the target strictly when the control is active.
- Primary Use: Generating Maximally Entangled States (Bell States), reversible computing, and logical dependencies.



CX State Transformations

Mathematical Mapping

The target qubit flips only when the control is 1.

$$
\begin{aligned}
|00\rangle &\rightarrow |00\rangle \\
|01\rangle &\rightarrow |01\rangle \\
|10\rangle &\rightarrow |11\rangle \\
|11\rangle &\rightarrow |10\rangle
\end{aligned}
$$

Truth Table

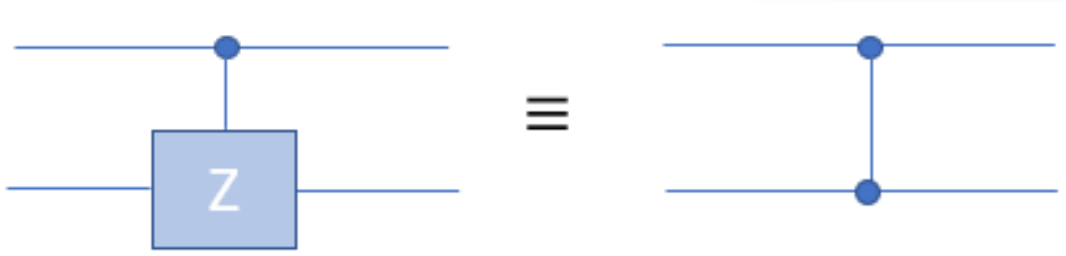
Classical representation of the quantum XOR.

Control (C)	Target (T)	Output State
0	0	$\|00\rangle$
0	1	$\|01\rangle$
1	0	$\|11\rangle$
1	1	$\|10\rangle$

2. CZ Gate (Controlled-Z)

The CZ gate diverges from the CX by conditionally applying a phase change rather than a physical state flip.

- Configuration: 1 control qubit, 1 target qubit.
- Mechanism: It applies a Z gate to the target strictly when the control is active, modifying only the phase.
- Primary Use: Quantum Fourier Transform (QFT), algorithms relying on interference, and creating cluster states.



$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

Manipulating Phase Information

$$|11\rangle \rightarrow -|11\rangle$$

Phase Inversion

The Unique Property of CZ

Because the CZ gate only activates when both qubits are set to 1, it isolates and modifies exactly one computational basis state.

The following states undergo no transformation:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle$$

This subtle manipulation of hidden phase information is critical for creating quantum interference patterns.

3. CCX Gate (Toffoli)

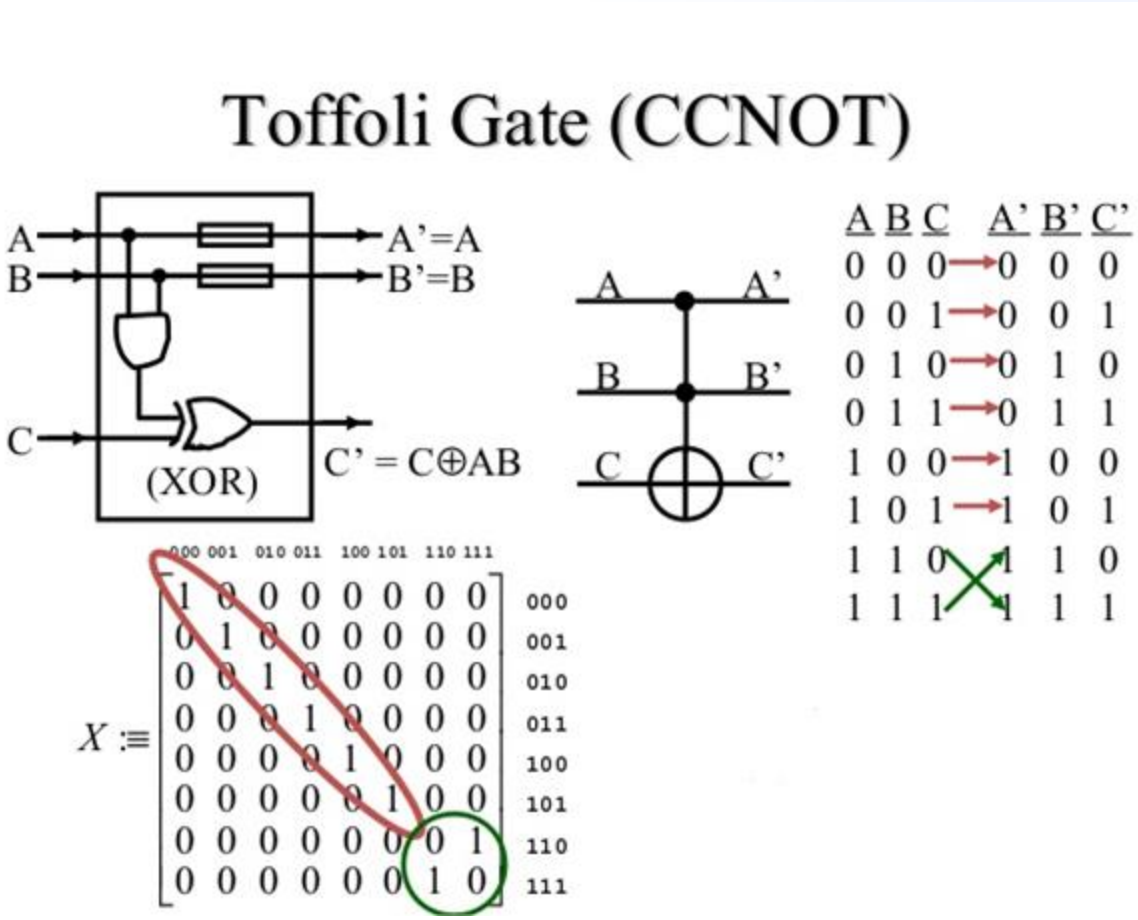
The Universal Reversible Gate

The Toffoli gate (CCNOT) is a three-qubit reversible logic gate with two control qubits and one target qubit. It flips the target bit strictly when both control bits are 1; otherwise, the target remains unchanged.

$$|a,b,c\rangle \rightarrow |a,b,c \oplus (a \wedge b)\rangle$$

Key Properties

- **Reversible:** Inputs can be fully recovered from outputs, preventing information loss.
- **Universal for Classical Logic:** Any reversible Boolean circuit can be constructed using only Toffoli gates.
- **Self-Inverse & Coherent:** Applying it twice restores the original state, and it acts on superpositions without collapse.



Source: https://anonymousket.medium.com/is-quantum-computing-so-different-e1fa2b1f90cb

CCX Gate Reversibility & Logic Simulation

Applying the CCX (Toffoli) gate twice sequentially perfectly restores the initial state, proving physical and mathematical reversibility.

INITIAL STATE

$|110\rangle$ (Controls: A=1, B=1 | Target: C=0)

↓ Apply CCX (Since both controls are 1, target flips)

INTERMEDIATE STATE

$|111\rangle$ (Target C successfully inverted to 1)

↓ Apply CCX Again (Controls remain 1, target flips back)

RECOVERED STATE

$|110\rangle$ (Original quantum state fully recovered)

Mathematical Key Point

Simulating Classical AND Reversibly

The Toffoli gate avoids information loss by mapping inputs to outputs via a reversible transformation instead of compressing them:

$$(A, B, C) \rightarrow (A, B, C \oplus A \cdot B)$$

When the target qubit is initialized to 0 (C = 0):

$$(A, B, 0) \rightarrow (A, B, A \wedge B)$$

This elegant mapping allows quantum hardware to simulate standard classical AND operations without losing state history.

Entanglement in Quantum Computing

A Simple Explanation of the Universe's Most Surprising Phenomenon

Entanglement is a quantum phenomenon where two or more quantum systems are connected such that their joint states cannot be described independently. As a result, their measurement results can be related to each other, even when they are far apart.

Measuring one part of an entangled system gives information about the measurement outcome of the other part.

Creating Entanglement

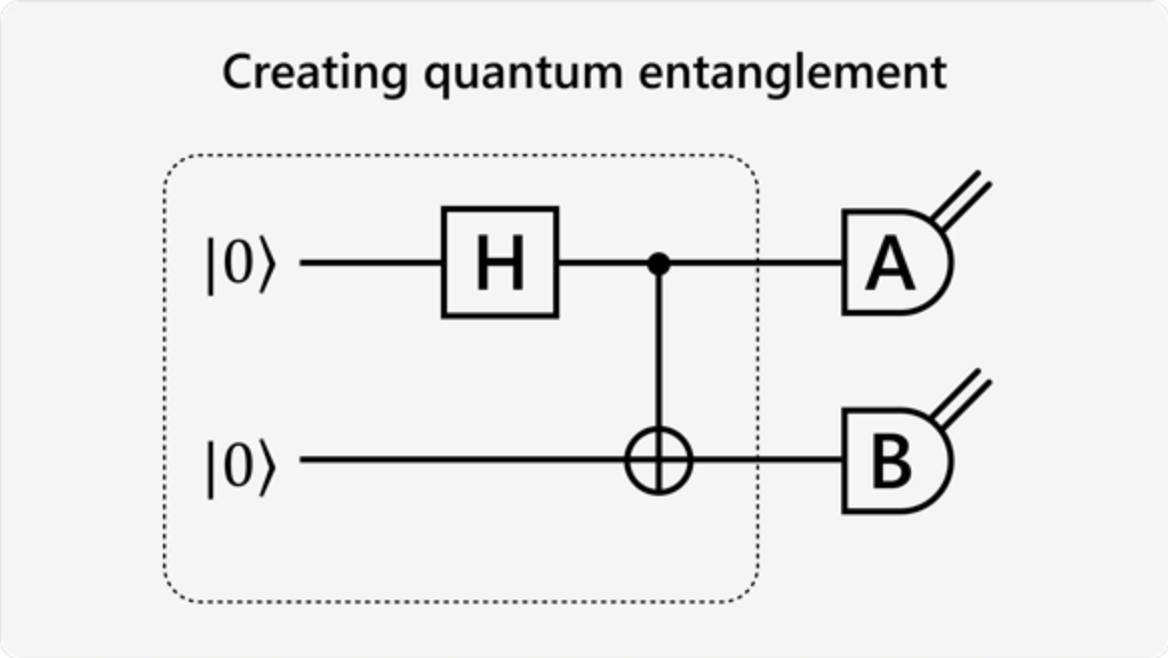
Start with two independent qubits in the zero state. To entangle them, we use a fundamental quantum circuit:

1. Hadamard Gate (H): Puts the first qubit into a superposition.

2. CNOT Gate (CX): Flips the target qubit only if the control qubit is 1.

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This creates one of the famous Bell States.



Understanding the CNOT Gate

The true power of the CNOT (Controlled-NOT) gate is its ability to take a superposition and create a joint state that can no longer be separated into independent qubit states.

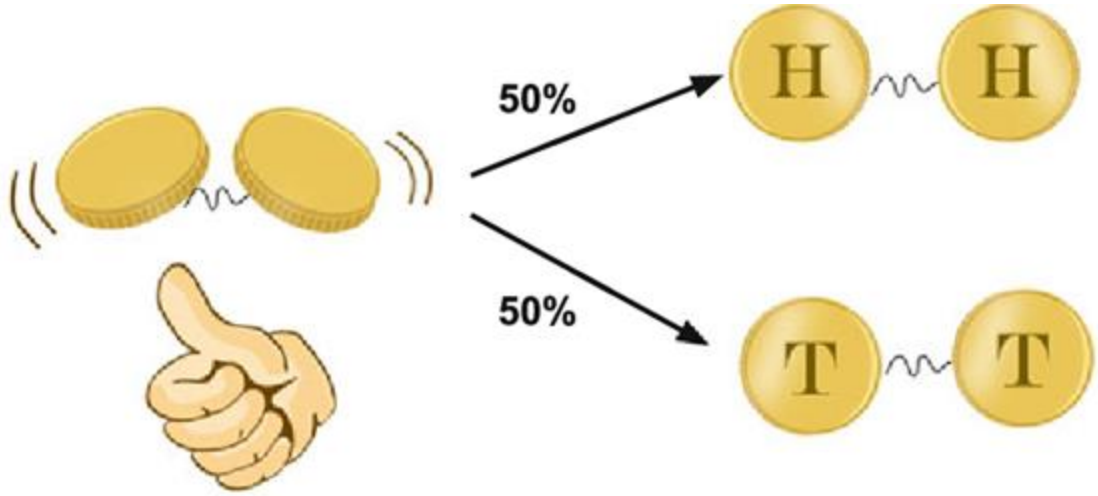
Control Qubit (A)	Target Qubit (B)	Output State
0	0	00 (No flip)
0	1	01 (No flip)
1	0	11 (Target flips)
1	1	10 (Target flips)

The Quantum Coin Analogy

Imagine two quantum coins prepared in an entangled state and separated across the world.

If you measure Coin A and it is Heads (0), you instantly know Coin B is also Heads (0). You will *never* get a mixed result (01 or 10).

Unlike classical coins secretly stamped before separation, quantum mechanics proves these correlations are stronger than any ordinary pre-existing values.



The Mathematical Signature

If a state is not entangled, you can mathematically separate it back into individual qubit states.

But for an entangled state, this separation is mathematically impossible. This non-separability is the strict definition of bipartite pure-state entanglement.

$$|\psi\rangle \neq |A\rangle \otimes |B\rangle$$



Defining Entanglement

Pure States

For a bipartite pure state, if the state cannot be written as a tensor product of two single-system states, it is entangled.

Separable State

$(|00\rangle + |01\rangle + |10\rangle + |11\rangle)/2$

Factors cleanly into $|+\rangle \otimes |+\rangle$

Entangled State (Bell State)

$|\psi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$

Mathematically impossible to factor into two independent states.

Mixed States

A bipartite mixed state ρ_{AB} is separable if it can be written as:

$$\rho_{AB} = \sum_i p_i \, \rho_A^{(i)} \otimes \rho_B^{(i)}$$

where $p_i \geq 0$ and $\sum p_i = 1$

If no such decomposition exists, the mixed state is considered **entangled**.

For pure states, we check whether the state vector can be **factored**.
For mixed states, we check whether the density matrix can be expressed as a **probabilistic mixture** of product states.

Importance for Algorithms

Entanglement enables quantum algorithms to create and exploit correlations between multiple qubits. It is an important quantum resource that can contribute to quantum advantage for certain problems. It plays an important role in:

- Quantum Simulation & Error Correction Modeling complex molecular structures and protecting quantum data.
- Quantum Teleportation Transmitting quantum information reliably between systems.
- Quantum Cryptography Securing communication channels using entangled states.
- Variational Quantum Algorithms Optimizing complex solutions in QAOA and VQE frameworks.
- Quantum Machine Learning (QML) Enabling rich, multi-qubit data interactions in neural networks.

Superposition says a single qubit can exist in a combination of possible states; entanglement says the states of multiple qubits can become inseparably connected.

Bell States

A Bell state is a specific quantum state involving two qubits that are maximally entangled.

Breaking Down the Notation

Understanding |00⟩

In quantum computing, a two-qubit state is written together. In the state |00⟩, the first number belongs to qubit 1 (q_1) and the second to qubit 0 (q_0).

This strictly means both qubits are in the classical zero state.

The Four Basis States

There are four possible computational-basis states for a two-qubit system:

$|00\rangle \rightarrow q_1 = 0, q_0 = 0$

$|01\rangle \rightarrow q_1 = 0, q_0 = 1$

$|10\rangle \rightarrow q_1 = 1, q_0 = 0$

$|11\rangle \rightarrow q_1 = 1, q_0 = 1$

The First Bell State: Φ+

Let's look at the most famous Bell state equation:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The fraction represents the **probability amplitude**.

It dictates the likelihood of obtaining specific outcomes upon measurement. The square of this amplitude gives the actual probability.

When measured, you will get:

|00⟩: 50% chance

|11⟩: 50% chance

We will always get matching results. That perfectly correlated relationship is entanglement.

Creating a Bell State

To create a maximally entangled state, we must execute a specific sequence using a Hadamard (H) gate and a Controlled-NOT (CNOT) gate.

The Quantum Circuit: Step-by-Step

1. Initialize

Start by setting both qubits strictly to the zero state.

$$|\psi\rangle = |00\rangle$$

2. Apply H Gate

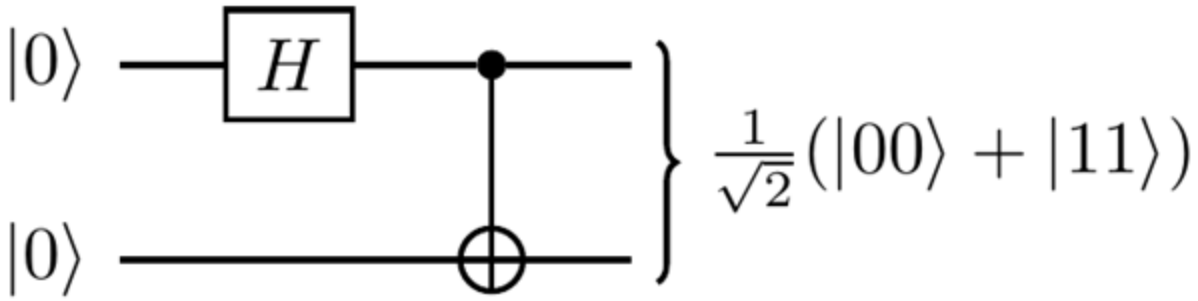
The Hadamard gate puts q0 into a superposition, leaving q1 as 0.

$$\rightarrow \frac{1}{\sqrt{2}}(\,|00\rangle + |10\rangle\,)$$

3. Apply CNOT

If the control (q0) is 1, flip target (q1). The $|10\rangle$ state becomes $|11\rangle$.

$$\rightarrow \frac{1}{\sqrt{2}}(\,|00\rangle + |11\rangle\,)$$



The Four Bell States Cheat Sheet

Bell State	Equation	Measurements	Relationship	Phase
$\Phi+$	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	00 or 11	Same	Positive (+)
$\Phi-$	$\frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$	00 or 11	Same	Negative (−)
$\Psi+$	$\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$	01 or 10	Different	Positive (+)
$\Psi-$	$\frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$	01 or 10	Different	Negative (−)

Circuits for the Four Bell States

First Bell State

$|\Phi^+\rangle$ State

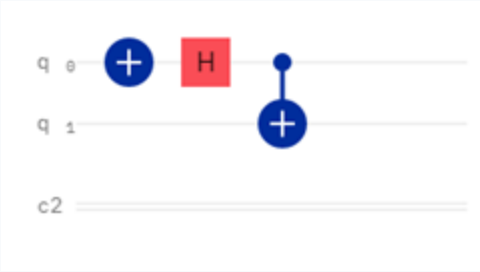


Configuration:

Standard circuit for implementing the first Bell state using a Hadamard gate followed by a CNOT gate.

Second Bell State

$|\Phi^-\rangle$ State

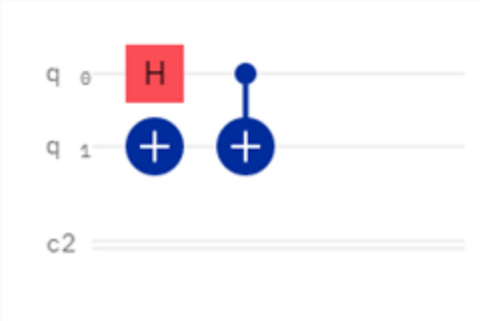


Configuration:

Circuit to implement the second Bell state, utilizing an initial X gate prior to the superposition stage.

Third Bell State

$|\Psi^+\rangle$ State

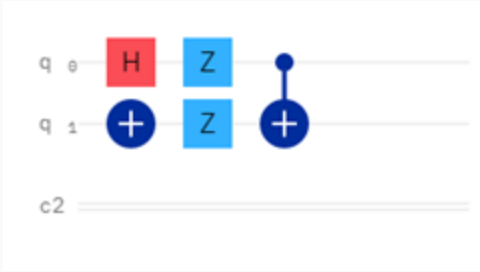


Configuration:

Circuit for the third Bell state, establishing anti-correlated entangled outcomes.

Fourth Bell State

$|\Psi^-\rangle$ State



Configuration:

Circuit for the fourth Bell state, applying phase shifts to obtain the singlet state.

Source: quantumcomputinguk.org/tutorials/introduction-to-bell-states

The Impact of Phase Difference



NEGATIVE PHASE

What does the minus sign mean?

You might wonder what the difference is between $\Phi+$ (plus sign) and $\Phi-$ (minus sign). That minus sign is a quantum phase difference.

It doesn't change the 50-50 probabilities when measuring directly in the standard computational basis. However, it completely changes how the qubits interfere with each other if you perform further operations or measure them from different angles.

Why Are Bell States Important?

Bell states are not just theoretical math problems; they are the fundamental building blocks of quantum architecture:

- Quantum Teleportation: Bell states create the entangled channel needed to teleport the state of an unknown qubit.
- Superdense Coding: A shared Bell pair allows two classical bits to be sent using only one transmitted qubit.
- Error Correction: Entanglement is vital for identifying and fixing errors in fault-tolerant computing.

Probability vs. Correlation

Individual Probability is Random

If you measure the Φ+ state, you have absolutely no way to predict whether you will get a 00 or an 11.

The *individual* outcome for either qubit is entirely random—a perfect 50/50 coin flip.

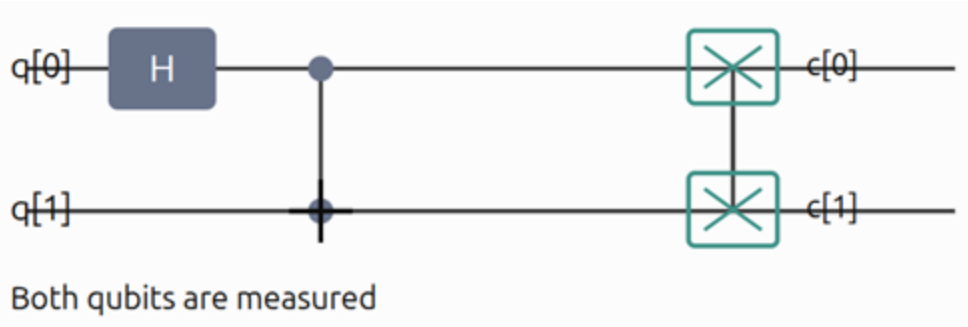
The Relationship is Correlated

However, the *relationship* between the two qubits is perfectly rigid. If q_0 is 0, q_1 is guaranteed to be 0.

We cannot predict the result, but we can definitively predict the connection between the results.



Only q[0] is measured



Both qubits are measured

Quantum Teleportation

Transmitting Unknown Quantum States via Classical Communication & Entanglement

The Problem Formulation

The Objective

Transfer a Quantum State

Alice holds a qubit in an unknown quantum state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The goal is to recreate this exact state in Bob's laboratory with absolute fidelity.

In an ideal noiseless protocol, Bob obtains exactly the unknown state. In real hardware, fidelity is less than 1 because of noise and imperfections.

The Core Constraint

No Physical Sending

Alice is strictly forbidden or physically unable to send the actual qubit containing the state to Bob.

Why not just measure and send the details?
Measurement collapses the superposition, destroying the state's information forever.

We must transmit the information without sending the physical carrier.

The Core Challenge

Teleportation explores transferring a state using strictly classical communication.

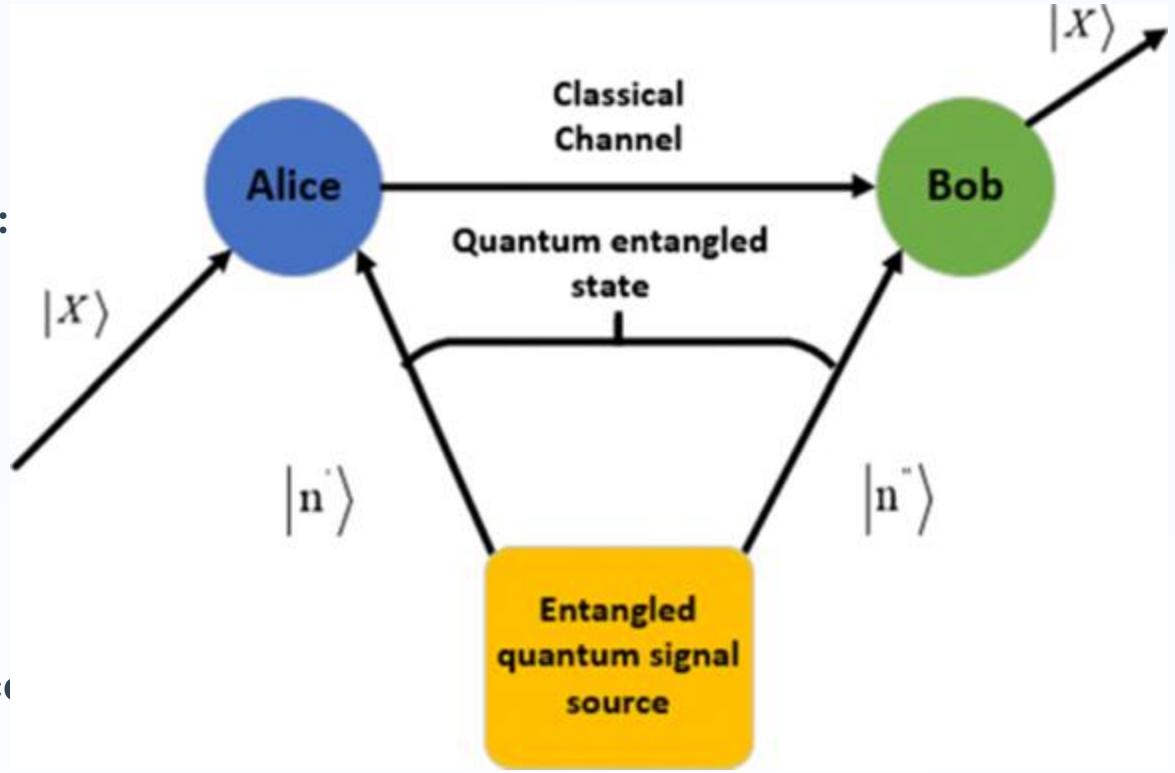
Imagine Alice holds a qubit in an unknown quantum state:

$$| \psi \rangle = \alpha | 0 \rangle + \beta | 1 \rangle$$

where the complex amplitudes satisfy the normalization condition:

$$| \alpha |^2 + | \beta |^2 = 1$$

Goal: Bob must receive a qubit in this exact state, but Alice cannot physically send her qubit to Bob.



The Flaws of Classical Solutions

Measurement Collapse

Why can't Alice simply measure and describe the state?

Alice is given an unknown quantum state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

A measurement yields only a classical outcome (0 or 1), collapsing the superposition.

Therefore, one measurement cannot reveal the complete state or its amplitudes.

No-Cloning & Complexity

The No-Cloning Theorem

This fundamental law of physics prevents Alice from making a perfect copy of an arbitrary unknown quantum state.

Determining the State

To estimate its amplitudes, Alice would need measurements on infinitely many identically prepared copies of the state (quantum state tomography), which she cannot clone.

How Teleportation Helps

Teleportation entirely avoids determining or transmitting amplitudes directly. By utilizing pre-existing shared entanglement and transmitting just **two classical bits**, the unknown state is reconstructed on Bob's qubit without ever collapsing or measuring the original carrier's quantum information.

The Quantum Alternative

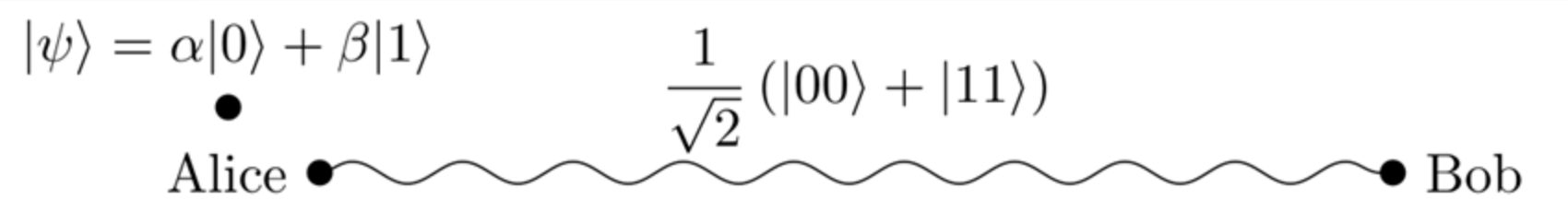
2

Classical Bits

Overcoming the Impossible

Quantum teleportation provides a brilliant workaround to the classical measurement problem.

If Alice and Bob share a pre-existing entangled pair, Alice can seamlessly transfer the unknown state to Bob by sending him just two classical bits, entirely bypassing the need to know the complex amplitudes.

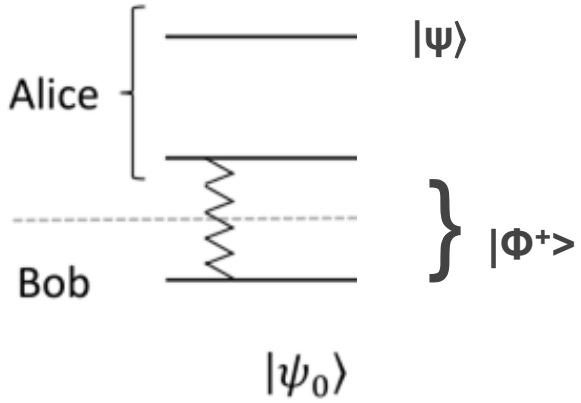


The Quantum Solution

A Step-by-Step Protocol Breakdown

The Three-Qubit Setup

Spatial Distribution



Physical Allocation

Alice holds Qubits 1 and 2 in her laboratory, while Qubit 3 is sent to Bob beforehand to establish the shared quantum channel.

Qubit 1

Alice's Unknown Qubit

The original quantum state that needs to be teleported.

Qubit 2

Alice's Entangled Half

The part of the shared Bell pair that resides with Alice.

Qubit 3

Bob's Entangled Half

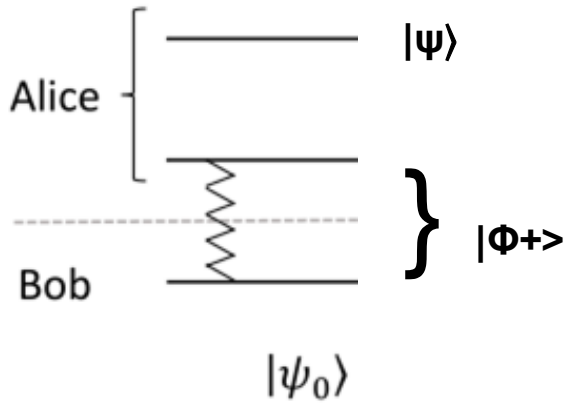
The part of the shared Bell pair residing with Bob. This receives the state.

Initial State Formulation

1. Shared Bell State

Alice and Bob share a standard Bell state prior to teleportation:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$



2. Combined Initial State

The combined initial state of all three qubits is formulated as the tensor product of the unknown state and the Bell state.

Combining and expanding the full state yields:

$$|\psi\rangle|\Phi^+\rangle = \frac{1}{\sqrt{2}}[\alpha(|000\rangle + |011\rangle) + \beta(|100\rangle + |111\rangle)]$$

Note: The first two qubits belong to Alice, and the third belongs to Bob.

This combined mathematical formulation serves as the baseline setup from which the quantum teleportation algorithm proceeds.

State Evolution: Alice's Operations

Step 1: Alice applies CNOT

Alice performs a **Controlled-NOT** operation on her two qubits. Qubit 1 is the control, Qubit 2 (entangled qubit) is the target.

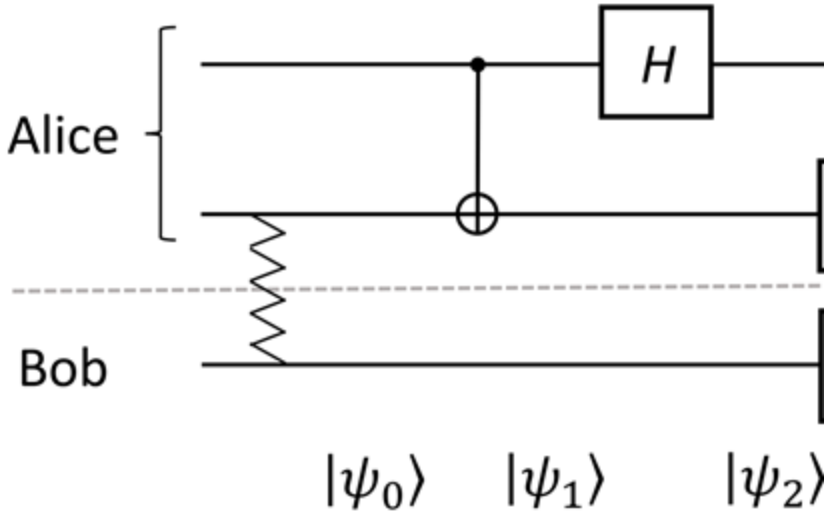
$|\psi_1\rangle$ $\frac{1}{\sqrt{2}}[\alpha(|000\rangle + |011\rangle) + \beta(|110\rangle + |101\rangle)]$

Step 2: Alice applies Hadamard

Next, Alice applies a **Hadamard (H)** gate to her first qubit. The three-qubit state expands to:

$\frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\beta|0\rangle + \alpha|1\rangle) + \ldots]$

Quantum Circuit & State Evolution



Combined State $|\psi_2\rangle$:

$|\psi_2\rangle$ = [1/2] [|00⟩ (α|0⟩+β|1⟩) + |01⟩ (β|0⟩+α|1⟩) + |10⟩ (α|0⟩−β|1⟩) + |11⟩ (−β|0⟩+α|1⟩)]

Step 3: Alice Measures

Alice measures both of her qubits in the computational basis. There are four equally likely measurement outcomes (each 1/4 probability). Bob's qubit collapses into one of four corresponding states:

Alice's Result	Bob's Resulting State	Required Correction	Meaning
00	$\alpha\|0\rangle + \beta\|1\rangle$	I (Identity)	Already the target state
01	$\beta\|0\rangle + \alpha\|1\rangle$	X (Bit flip)	Amplitudes exchanged
10	$\alpha\|0\rangle - \beta\|1\rangle$	Z (Phase flip)	Relative phase changed
11	$-\beta\|0\rangle + \alpha\|1\rangle$	XZ (Both) [since XZ = -ZX, differ only by a global phase and represent the same physical state]	Combined bit/phase transformation

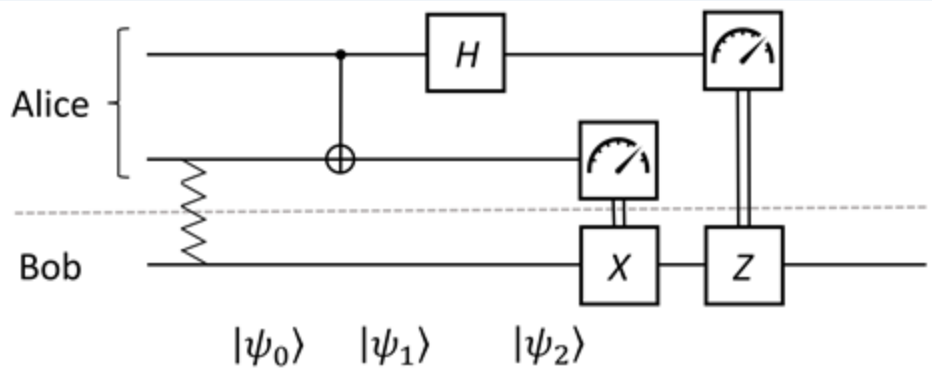
Steps 4 & 5: Classical Communication & Correction

Step 4: Classical Communication

Alice must now inform Bob of her measurement results via a standard classical communication channel.

She simply transmits the two bits she measured (00, 01, 10, or 11).

This is the only information that physically traverses space between them.



Step 5: Applying Corrections

Alice sends her 2 measurement bits to Bob. Based on them, Bob applies the required gate to recover the original state $|\psi\rangle$.

00 → Do nothing I
01 → Apply X (to flip bits)
10 → Apply Z (to flip phase)
11 → Apply X, then Z

After the correction:

$$\text{Bob's qubit} = |\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Result: Bob successfully restores the exact state.

In simple words: Alice's two classical bits tell Bob exactly which quantum correction to apply, enabling perfect state recovery.

The Complete Protocol Sequence

STEP 01

Unknown State

The initial input qubit $|\psi\rangle$ that needs to be teleported to Bob.

STEP 02

Entanglement

Creation of a shared maximally entangled Bell pair between Alice & Bob.

STEP 03

CNOT Gate

Alice performs a joint CNOT operation on her two qubits.

STEP 04

Hadamard (H)

Alice applies an H gate to change the basis of the first qubit.

STEP 05

Measurement

Alice measures both of her qubits in the computational basis.

STEP 06

2 Classical Bits

The 2-bit measurement result is transmitted to Bob classically.

STEP 07

Correction

Bob applies recovery operators (I, X, Z, or XZ) to his qubit.

STEP 08

Recovered State

Bob successfully reconstructs the exact original state $|\psi\rangle$.

What Actually Gets Teleported?

A common misconception is that a physical particle travels from Alice to Bob. **This does not happen.**

Teleportation transfers the unknown quantum state from Alice's qubit to Bob's qubit; the physical qubit itself does not travel from Alice to Bob.

Initially: Alice's qubit holds the state.

During the protocol: Alice performs measurements on her qubits and sends two classical bits to Bob.

After the protocol: Bob applies the required correction, and Bob's qubit holds the original quantum state.

No-Cloning Theorem Consistency

The original state is no longer available on Alice's qubit, so teleportation does not create a second copy. This is consistent with the no-cloning theorem.

State Transfer Mechanism

The qubit does not travel — its quantum state is transferred to Bob's qubit using entanglement and classical communication.

One-Line Intuition

> Quantum teleportation uses shared entanglement and two classical bits to perfectly transfer an unknown quantum state from Alice to Bob, entirely without sending the original physical particle.

Thank you!